Personality and Individual Differences 172 (2021) 110596

Contents lists available at ScienceDirect

Personality and Individual Differences

journal homepage: www.elsevier.com/locate/paid

Review

Personality development from early childhood through adolescence

Helena R. Slobodskaya *

Scientific Research Institute of Physiology and Basic Medicine, Novosibirsk, Russia
Novosibirsk State University, Russia

ARTICLE INFO

Keywords:
Personality development
Childhood
Adolescence
Gender differences
Mean-level age differences

ABSTRACT

Recent years have seen a great increase in research on personality development; however, most research has employed self-report questionnaires and concerned individuals older than 10 years. Whereas individual differences in younger children have traditionally been studied as temperament, studies have begun to explore personality development in the first ten years of life using parent and teacher ratings. The Big Five traits of Extraversion, Neuroticism, Agreeableness, Conscientiousness and Openness have accumulated the most evidence on age changes in personality; relatively less is known about age differences in lower-order traits, and evidence on the development of the higher-order traits is lacking. In this article, I briefly describe the hierarchical structure of child and adolescent personality, summarise research on developmental trends in mean levels of personality traits across the first ten years of life and in adolescence and address gender differences in the development of traits. I conclude by highlighting some directions for future research. The scope of the present review is limited to normal personality traits in childhood and adolescence measured by widely used instruments.

1. Introduction

The last two decades have seen a great increase in studies of personality development; the findings have been summarised in several volumes (McAdams et al., 2019; Mroczek & Little, 2006; Specht, 2017) and reviews (Caspi et al., 2005; Costa Jr et al., 2019; McAdams & Olson, 2010). So far, however, most research has employed self-report questionnaires and has concerned individuals older than ten years, the age at which children are able to reliably rate themselves (Costa Jr et al., 2019; Soto et al., 2011). Individual differences in younger children have traditionally been described as temperament, and there has been little contact between researchers in these two fields. It is increasingly recognised that personality and temperament traits are more alike than different (Caspi et al., 2005; Shiner & DeYoung, 2013). In recent years, studies have begun to explore personality development in the first ten years of life using parent and teacher ratings. In this article, I briefly describe the hierarchical structure of child and adolescent personality, summarise research on developmental trends in mean levels of personality traits in childhood and adolescence, address gender differences in the development of traits and suggest some directions for future research. The scope of the present review is limited to normal personality traits in childhood and adolescence measured by widely used instruments.

2. The hierarchy of personality traits in childhood and adolescence

In the past three decades, researchers largely agreed about the hierarchical structure of adult personality (Markon et al., 2005). The Big Five traits, Extraversion, Neuroticism, Agreeableness, Conscientiousness and Openness, subsume more specific, lower-order ones and represent the foundational level of superordinate personality hierarchy from which the other Big Trait models can be derived. The general consensus among researchers of adult personality has encouraged studies on the structure of individual differences in childhood and adolescence (Halverson et al., 1994; Kohnstamm et al., 1998). This section provides a brief overview of evidence on the hierarchical structure of child and adolescent traits in a cross-cultural perspective.

2.1. The top-down approach to the trait hierarchy

Structural studies with children and adolescents have generally

* Department of Child Development and Individual Differences, Scientific Research Institute of Physiology and Basic Medicine, Timakova str., 4, Novosibirsk 630117, Russia.

E-mail address: hslob@physiol.ru.

https://doi.org/10.1016/j.paid.2020.110596
Received 4 June 2020; Received in revised form 10 December 2020; Accepted 11 December 2020
Available online 25 December 2020
0191-8869/© 2020 Elsevier Ltd. All rights reserved.

supported the five-factor structure when using a top-down approach which relies on personality models and measures developed for adults (De Pauw, 2017). Thus, a large-scale study using college student observer ratings of adolescents from 24 cultures on an adult personality measure, found that the five-factor structure of personality was nearly identical in adolescents aged 12–14 and 15–17 years (De Fruyt et al., 2009). Still, there were minor differences; for example, Openness was not replicated in adolescents from seven African, Asian, European and South American countries although this trait was clearly identified in most of these countries in adults.

Structural research with parent and teacher ratings on child-specific measures of personality characteristics not specifically designed to assess the Big Five traits has also supported the five-factor structure that has been repeatedly found in adults (De Pauw, 2017; Halverson et al., 1994). Still, the utility of the five-factor model for understanding early development has not been without criticism (Block, 2010). A large-scale study of parent reports on the broad set of personality-focused descriptions in English language (Soto & John, 2014) found that the one-, two- and three-factor structures replicated in each age group from 3 to 18–20 years of age. The general factor, labeled Adjustment, broke down into Benevolence and Emotionality, which bifurcated into Positive and Negative Emotionality at the three-factor level. The four-factor structure began to replicate from age 7; the five-factor structure also became more replicable with age. This study suggested that the Little Six dimensions, including the Big Five and Activity, may constitute the foundational level of superordinate personality hierarchy from middle childhood through adolescence (Soto & John, 2014). However, a top-down approach may overlook age-specific or indigenous dimensions.

2.2. The bottom-up approach

A bottom-up approach to personality structure derives evidence for traits from factor analyses of the representative set of empirical descriptions of individual differences (Cheung et al., 2011; Halverson et al., 2003). Although this approach is invaluable in revealing similarities and differences in personality structure across ages, it has been most commonly used in cross-cultural structural research. Lexical studies use trait descriptors extracted from dictionaries to examine personality structure in different languages. Cross-language studies showed that whereas the Big Five is clearly replicable in European-American languages, one-, two- and three-factor structures are more generalisable across the world (De Raad & Mlačič, 2017). The Big One, or the General Factor of Personality, measures desirable vs undesirable characteristics, and is still the subject of much debate (Just, 2011; van der Linden et al., 2013). The Big Two are very similar to the higher-order factors of the Big Five, Alpha comprising Agreeableness, Conscientiousness and Neuroticism (reversed), and Beta comprising Extraversion and Openness, that have been consistently recovered in studies with children, adolescents and adults using a top-down approach (Shiner & DeYoung, 2013). The three-factor structure, including Extraversion, Agreeableness, and Conscientiousness, proved highly replicable across languages (De Raad & Mlačič, 2017). Overall, the findings of lexical studies are likely to be relevant for understanding the structure of child and adolescent traits. Research that has sought to examine the validity of the five-factor model for describing the structure of free parental descriptions of child and adolescent personality is discussed next.

2.3. Combining approaches

Probably the best way to understand personality structure across ages and cultures is to combine the methodological rigor of the top-down approach with the sensitivity of the bottom-up approach (Cheung et al., 2011). Only one published study to date used this strategy to examine child and adolescent personality. An international research team collected over 50,000 free parental descriptions of 2–14-year-old children in seven cultures, Belgium, China, Germany, Greece, the Netherlands, Poland and the United States (Kohnstamm et al., 1998). These descriptions were then coded into the Big Five traits: Agreeableness (considerate, manageable), Conscientiousness (organised, achievement oriented, good concentration), Neuroticism (fearful, insecure, quick-tempered), Extraversion (active, optimistic, sociable), and Openness (curious, intelligent), along with additional categories included on the empirical or rational grounds. Overall, more than 80% of the parental descriptions were classified into the Big Five categories (Halverson et al., 2003).

The subsequent analyses of age-specific indigenous instruments resulted in a culturally and age decentered set of personality descriptors that matched the distribution of the Big Five categories in the parental lexicon to measure fifteen specific (or lower-order, fine-grained) traits that formed the Big Five (Halverson et al., 2003). A study of this personality measure, the Inventory of Child Individual Differences (ICID), in two Slavic counties, Slovenia and Russia, established full measurement invariance of the five-factor structure across cultures, parent and self-reports and ages 2 to 15 years (Knyazev et al., 2008). The work with Flemish parental descriptions of child personality across ages 6–13 years confirmed the five-factor structure similar to the adult Big Five: Extraversion, Conscientiousness, Emotional Stability (Neuroticism reversed), Benevolence (Agreeableness) and Imagination (Openness), and led to the development of the Hierarchical Personality Inventory for Children (HiPIC), consisting of 18 highly reliable facets (Mervielde & De Fruyt, 1999). Later, the HiPIC factor structure has been recovered in parent ratings of preschool children as well as in parent and self-ratings of adolescents (De Pauw, 2017). An important strength of the ICID and HiPIC is that both instruments are based on bottom-up collection of free parental descriptions and analyses of age-specific personality structures.

Tackett et al. (2012) used the ICID item set to examine the hierarchical structure of child and adolescent personality in five countries: Canada, China, Greece, Russia and the United States. The findings showed that for all countries and age groups from 2 to 14 years, the Big Traits and their hierarchical relationships were very similar to established patterns in the adult literature (see Fig. 1). The three-factor level closely resembled well-established models of child temperament, with the three overarching factors of Positive Emotionality, Negative Emotionality and Effortful Control/Inhibition (Rothbart, 2011). Thus, the hierarchical structure of individual differences provides a bridge between temperament and personality, although their relations remain to be investigated (De Pauw, 2017). In this and other studies, there were notable age differences in the hierarchical relationships and the content of traits, highlighting important directions for future research on child and adolescent traits.

Overall, research on personality structure across ages and cultures suggests that the structural hierarchy of traits is broadly universal, allowing for the examination of measurement invariance and comparisons of mean trait levels at different ages. The next section reviews the literature on developmental trends in levels of personality traits across childhood and adolescence. The following criteria were used to select studies: (a) English language; (b) studies of children and/or adolescents; (c) community samples; (d) personality measures based on the five-factor framework; (e) comparisons of mean trait levels across ages. It was beyond the scope of this review to consider studies of maladaptive traits or problem behaviours, although they may also provide valuable information on the development of individual differences.

3. Age differences in child and adolescent personality traits

At all ages, traits show both stability and change over development. Rank-order or differential stability reflects the maintenance of a person's relative standing within a group over time. Longitudinal research shows that rank-order stability is moderate in childhood, increases during adolescence and early adulthood and does not differ substantially across traits and genders (Caspi et al., 2005; Costa Jr et al., 2019). Rank-order stability coexists with mean-level change, which refers to age-related

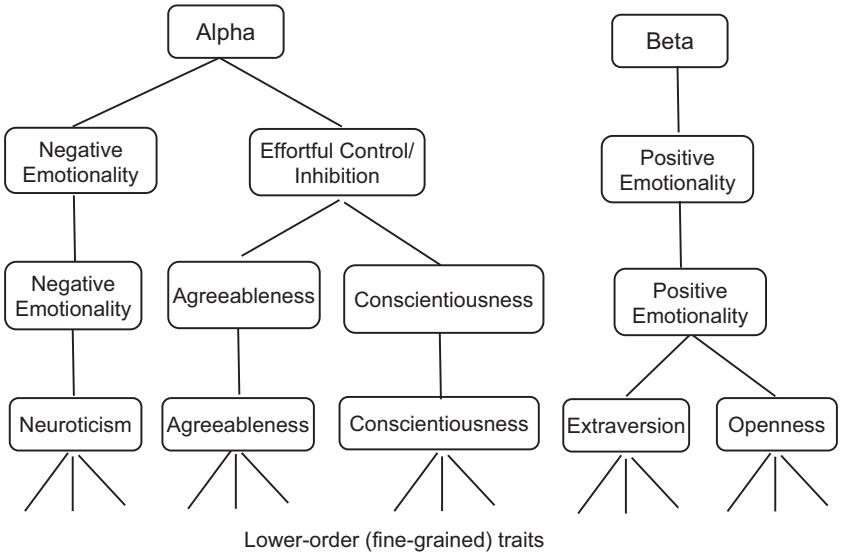

Fig. 1. Hierarchical structure of individual differences in childhood.

differences in the average level of a given trait in a group. In contrast to rank-order stability, mean-level changes in personality differ markedly across traits and, sometimes, genders (De Haan et al., 2017; Klimstra et al., 2009; Pullmann et al., 2006; Zupančič et al., 2009). Moreover, mean-level change in personality traits does not imply that all children have the same developmental trajectories: there are individual differences in the amount and direction of change (Borghuis et al., 2017; Branje et al., 2007). However, knowledge of normative age trends of trait levels is crucial for understanding personality development (Costa Jr et al., 2019). In the following sections, evidence about mean-level changes in personality traits in childhood and adolescence is organised by the higher-order factors of the Big Five, Alpha comprising Agreeableness, Conscientiousness and Neuroticism (reversed), and Beta comprising Extraversion and Openness.

3.1. Age differences in mean levels of childhood traits

Because most research on child individual differences has been conducted in the field of temperament (De Pauw, 2017; Halverson et al., 1994; Shiner & DeYoung, 2013), only a few studies have examined personality development in the first ten years of life. Both longitudinal and cross-sectional studies suggest that from early childhood through elementary school years, children appear to become more agreeable. This was evidenced by increases in mean level of Agreeableness (Benevolence) and the lower-order traits of altruism and compliance and by decreases in strong willed, egocentrism and dominance reported by parents (De Haan et al., 2017; Laidra et al., 2017; Lamb et al., 2002; Van den Akker et al., 2014) and teachers (Prinzie & Deković, 2008). The study of free parental descriptions of 2–13-year-olds in seven countries found that parents of older children more frequently used descriptors referring to high Agreeableness (e.g., caring, well behaved, sincere) and less frequently used low Agreeableness descriptors (e.g., selfish, stubborn, lies) compared with parents of younger children (Slotboom et al., 1998). However, some studies did not find this positive trend for Agreeableness (Soto, 2016).

Evidence from four European longitudinal studies suggests an increase in Conscientiousness from early childhood and preschool years to elementary school years, but no age differences or a decrease from middle childhood into adolescence; the lower-order traits of achievement, concentration and organisation showed a similar pattern (De Haan et al., 2017; Laidra et al., 2017; Lamb et al., 2002; Zupančič et al., 2009). Other cross-sectional and longitudinal studies also suggest that from middle childhood into adolescence children appear to become less conscientious (Soto, 2016; Van den Akker et al., 2014) and achievement motivated (Prinzie & Deković, 2008). The study of free parental descriptions found that, in seven widely different countries, parents of 3-year-old children less frequently used descriptors referring to high Conscientiousness (e.g., good concentration, reliable, hardworking) than parents of 6- and 9-year-old children; however, the use of descriptors referring to low Conscientiousness (e.g., forgetful, careless, lazy) showed similar age trends (Slotboom et al., 1998). Overall, the increases in mean levels of Agreeableness and Conscientiousness in the first years of life are consistent with the development of Effortful Control (Rothbart, 2011) and prosocial behaviour (Eisenberg et al., 2015), and with the well-documented decreases in physical aggression and behaviour problems (Tremblay, 2010).

Most studies suggest that children appear to become less emotionally stable from early childhood into middle childhood. This was evidenced by increases in mean level of Neuroticism (Laidra et al., 2017; Lamb et al., 2002; Soto, 2016; Van den Akker et al., 2014) and lower-order traits of anxiety and shyness and by decreases in the trait of self-confidence (De Haan et al., 2017; Prinzie & Deković, 2008; Slobodskaya & Akhmetova, 2010). An obvious explanation is that children become increasingly aware of the environmental demands they may have difficulty meeting (De Haan et al., 2017), and this may contribute to an increase in their levels of distress.

There is very consistent evidence that mean levels of Extraversion and the traits of sociability, positive emotions and activity appear to decrease across childhood. The findings for Openness are similar but less consistent, with most studies reporting higher levels of Openness and lower-order traits from this domain in younger children relative to older children and adolescents (De Haan et al., 2017; Laidra et al., 2017; Lamb et al., 2002; Prinzie & Deković, 2008; Slobodskaya & Akhmetova, 2010; Slotboom et al., 1998; Soto, 2016; Van den Akker et al., 2014) and some reporting increases in Openness (Imagination) and the trait of Intellect in early childhood and preschool years (De Haan et al., 2017; Laidra et al., 2017; Lamb et al., 2002). Together, these findings suggest that levels of child's exploration and engagement with the world encompassed by the Beta traits (Shiner & DeYoung, 2013) are highest in early childhood and gradually decrease through preschool and elementary school years.

Overall, mean-level changes in personality traits across the first ten years of life resemble well-documented maturational changes in adult personality – Agreeableness and Conscientiousness increase and Extraversion and Openness decline (Costa Jr et al., 2019); with the exception of Neuroticism, which tends to increase during childhood but declines in

adulthood. Evidence on developmental trends in levels of personality traits across adolescence is reviewed next.

3.2. Age differences in mean levels of adolescent traits

Research on personality development in adolescence has generated the disruption hypothesis (Denissen et al., 2013; Soto, 2016), suggesting that, on average, transition from childhood to adolescence is accompanied by temporary dips in personality maturation. Supporting this hypothesis, studies have found a U-shaped age trend for Agreeableness, with mean levels decreasing from late childhood into early adolescence and increasing from late adolescence into early adulthood (Soto, 2016; Soto et al., 2011; Van den Akker et al., 2014). A meta-analysis of age differences in mean levels of the Big Five across adolescence revealed a similar pattern of age differences for Conscientiousness and Openness (Denissen et al., 2013). Some studies report evidence for an inverted U-shaped age trend for Neuroticism with an increase in early and middle adolescence and a decrease in late adolescence and early adulthood; although the findings are not consistent across gender and informant (Borghuis et al., 2017; De Haan et al., 2017; Soto et al., 2011).

At the same time, other evidence seems to show personality maturation during adolescence, with mean levels of Agreeableness, Conscientiousness and Openness increasing and mean level of Neuroticism declining (Klimstra et al., 2009; Luan et al., 2017). The findings for Extraversion are also inconsistent: some studies found a decrease in adolescence (Kawamoto & Endo, 2015; Van den Akker et al., 2014), others found an increase (Klimstra et al., 2009; Pullmann et al., 2006), and still others showed no mean-level changes (Luan et al., 2017).

It is important to note that adolescents and their parents may vary in the direction and the magnitude of reported age changes, as well as the traits in which they observe the change (Göllner et al., 2017; Luan et al., 2017). For example, in a longitudinal study of 12- to 18-year-olds, adolescents reported increases in mean levels of Agreeableness, Conscientiousness and Openness, and no change in Neuroticism, their siblings' ratings showed a similar pattern; their fathers' and mothers' ratings, by contrast, showed declines in mean levels of Agreeableness, Openness and Neuroticism and no change in Conscientiousness (Luan et al., 2017). These differences may be due to contextual specificity of trait-related behaviour, to trait visibility to outside observers and to informants' unique perspectives on personality maturation. There is also evidence that perceiver effects represent a trait-like tendency toward positivity of judgements (Wood et al., 2010). Finally, gender differences may also have a role (Göllner et al., 2017). The next section summarises the evidence on gender differences in child and adolescent traits.

4. Gender differences in personality traits across childhood and adolescence

4.1. Gender differences in mean levels of traits

There is accumulating evidence about gender differences in personality/temperament from early childhood onwards, with girls, on average, scoring higher on Agreeableness, Conscientiousness and Effortful Control, and boys, on average, scoring higher on Activity (De Fruyt et al., 1998; Else-Quest et al., 2006; Soto, 2016; Van den Akker et al., 2014; Zupančič et al., 2008). Still, the findings were not entirely consistent; some studies have found little or no gender differences in mean levels of traits at younger ages (De Haan et al., 2017; Laidra et al., 2017; Lamb et al., 2002). One possible explanation for this discrepancy is that the magnitude of gender differences in child personality varies across cultures (Lamb et al., 2002). However, in a study of parental descriptions of 2–13-year-olds gender differences were similar across seven countries (De Fruyt et al., 1998).

In a large-scale study of adolescents, the direction of gender differences in observer reports was fairly uniform across 23 cultures from Africa, North and South America, Asia and Europe: girls, on average, scored higher than boys on all Big Five personality traits and most lower-order traits (De Bolle et al., 2015). The magnitude of these differences was small, gender accounted for 1.4% of the variance in the Big Five, whereas culture accounted for 3.5% (McCrae et al., 2010). It is also notable that the patterns and magnitude of gender differences may depend on the informant. For example, gender differences in mother and father ratings of 3 to 5-year-olds were larger than in preschool teacher reports (Zupančič & Kavčič, 2005). Whereas the effects of culture and informant on gender differences in child and adolescent traits await further study, the possibility that gender differences in traits may be evident from very early in development raises at an important question: does personality change in childhood and adolescence differ by gender? Studies that have addressed this question are reviewed next.

4.2. Gender differences in the development of traits

There is at present very little evidence about gender differences in personality development across the first ten years of life. Many cross-sectional and longitudinal studies have reported significant age-by-gender interactions for the Big Five and lower-order traits in adolescence, however, again, the results are equivocal. In some studies, mean levels of Agreeableness and Conscientiousness traits increased or remained stable across adolescence in girls and showed a curvilinear U-shaped age trend in boys, suggesting that personality maturation occurs earlier in girls (Borghuis et al., 2017; Klimstra et al., 2009; Slobodskaya & Akhmetova, 2010). In a longitudinal Flemish study, the shape of mean-level changes in the Agreeableness and Conscientiousness traits differed between boys and girls, however, these age trends were difficult to interpret (De Haan et al., 2017). Other studies of adolescents have found no evidence for gender differences in developmental trends for Agreeableness and Conscientiousness (Göllner et al., 2017; Pullmann et al., 2006).

Many studies have shown that mean level of Neuroticism appears to increase in girls and remains stable or decreases in boys during adolescence (Borghuis et al., 2017; De Haan et al., 2017; McCrae et al., 2002; Soto, 2016; Soto et al., 2011). However, in some studies girls' Neuroticism tended to decrease or remained stable in adolescence (Göllner et al., 2017; Klimstra et al., 2009; Pullmann et al., 2006) and boys' Neuroticism tended to increase (Kawamoto & Endo, 2015; Van den Akker et al., 2014). In a large-scale cross-cultural study of adolescents, both boys' and girls' Neuroticism tended to decline with age, but this effect in boys was stronger than in girls leading to well-documented gender differences in adult Neuroticism (De Bolle et al., 2015). Overall, gender differences in Neuroticism and temperamental Negative affectivity are almost negligible in childhood (De Fruyt et al., 1998; Else-Quest et al., 2006; Soto, 2016; Van den Akker et al., 2014), appear and increase in magnitude during adolescence, leading to well-documented gender differences in adult Neuroticism (Borghuis et al., 2017; De Bolle et al., 2015; Klimstra et al., 2009;Soto et al., 2011; Van den Akker et al., 2014). It should be noted, however, that the findings for children were obtained with parent reports, while the findings for adolescents were obtained with self-reports.

Some studies found that age-related decreases in Extraversion were more prominent in boys than in girls (Soto et al., 2011; Van den Akker et al., 2014). In other studies of adolescents, mean levels of Extraversion tended to increase among girls, but not boys (Branje et al., 2007; Göllner et al., 2017). The findings for the lower-order traits of optimism (De Haan et al., 2017) and assertiveness (De Bolle et al., 2015) were also inconsistent. There is some evidence that age trend for Openness from late childhood into late adolescence may differ by gender. One large cross-sectional study using youths' self-reports from English-speaking countries found a decline among girls and a curvilinear U-shaped pattern among boys (Soto et al., 2011), whereas in a longitudinal Dutch study self-reported Openness showed an inverted U-shaped trend among girls and increased linearly among boys (Borghuis et al., 2017). A large cross-sectional study using parent reports from English-speaking

countries found a curvilinear U-shaped age trend among girls and a decline among boys (Soto, 2016), whereas in a Russian study parent-reported Openness did not change among girls and showed an U-shaped age trend among boys (Slobodskaya & Akhmetova, 2010). Interestingly, a longitudinal multi-informant study of adolescents found that mean level of self-reported Openness increased among girls and did not change among boys, whereas mean level of parent-reported Openness did not change among girls and decreased among boys (Branje et al., 2007). The discrepancy between results for Openness and Neuroticism obtained with parent reports and adolescent self-reports is consistent with the notion that more evaluative and less visible traits show higher informant disagreement (Luan et al., 2017).

Overall, the findings suggest that personality maturation occurs earlier in girls and that girls, on average, tend to score increasingly higher on Neuroticism than boys during adolescence. However, the literature is equivocal and the question remains whether and to what extent the development of child and adolescent traits differs by gender. Research into biological and environmental sources of personality change may help to clarify the effect of gender on the development of traits. The next section provides a brief overview of the processes underlying personality development in childhood and adolescence.

5. Underlying processes

Age changes in personality traits result from the dynamic interaction between person and environment. A meta-theoretical framework can incorporate biological and contextual theories providing general principles and processes for an integrative model of personality development. Thus, a bioecological model of human development proposes:

The form, power, content, and direction of the proximal processes affecting development vary systematically as a joint function of the characteristics of the *developing person,* the *environment* – both immediate and more remote – in which the processes are taking place, the nature of the *developmental outcomes* under consideration, and the social continuities and changes occurring over time during the historical period through which the person has lived (Bronfenbrenner, 1999, p. 5).

A dynamic systems approach can lead to deeper understanding of the mechanism and course of development by explicitly modelling the processes of temporal change of the system from the current state toward the next state (van Geert & Steenbeek, 2005). Life span developmental theory views personality as a dynamic system, employs a structural and a process-oriented approach to the study of the commonalities and interindividual differences in personality development along with intraindividual variability in behaviour taking into account the biological, psychological, social and environmental factors and mechanisms (Baltes et al., 2006). Baumert and colleagues (Allik et al., 2017; Baumert et al., 2017) provided a comprehensive and integrative account of the current state of knowledge about personality structure, personality development and psychological processes and mechanisms that explain behaviour in context.

The field of developmental psychology has much to offer to our understanding of child and adolescent personality. Thus, the idea of increasing differentiation across development should be helpful in interpreting the hierarchical relationships between personality traits, such as splitting of Effortful Control/Inhibition into Agreeableness and Conscientiousness. While it is useful to recognise that the trait hierarchy "provides possible levels of description, each of which becomes more or less appropriate depending on the theoretical and empirical context" (Markon et al., 2005, p. 153), it is also useful to examine age trends in the differentiation of personality traits. Studies have found that personality traits become increasingly differentiated with age: self-reported Big Five traits are less correlated at older ages across adolescence, with the largest age difference found for the correlation between Agreeableness and Conscientiousness (Allik et al., 2004; Soto et al., 2008). Structural research with parent reports also showed the differentiation of Agreeableness and Conscientiousness from early childhood to early adolescence (Tackett et al., 2012). Overall, the findings support the idea of increasing differentiation of traits across childhood and adolescence (Brandes et al., 2020; De Pauw, 2017; Shiner & DeYoung, 2013), although there are exceptions. For example, activity level emerges as a distinct, basic dimension of temperament/personality in childhood (De Pauw et al., 2009), but then gradually merges with Extraversion (Soto & John, 2014).

Development of temperament in childhood and adolescence and underlying reactive and regulatory processes have been extensively studied (Rothbart, 2011). However, only a few studies have examined possible predictors of mean-level change in child and adolescent traits. One reason for this could be the age specificity of widely-used temperament measures. Another reason may lie in this field's emphasis on the stability of personality/temperament traits. Still, there is consistent evidence that parenting and family environment predict age changes in child Conscientiousness, Effortful Control and Negative Emotionality; findings for other traits are scarce (Lengua et al., 2019; Van den Akker et al., 2014). From adolescence onwards, identity development contributes to personality stability and change (Klimstra, 2013).

Denissen et al. (2013) suggested that personality development is driven by regulatory mechanisms and that disruption in personality maturation in adolescence may be due to temporal discrepancies between reference values and actual behaviour. This model emphasises deliberate personality development and has not been tested in childhood. However, regulatory mechanisms are involved in the development of temperament in the first years of life (Rothbart, 2011). The interplay of environmental influences and internal developmental processes may be especially important in accounting for normative trends and individual differences in child and adolescent trait change.

Distinctively different developmental trends for some traits within a higher-order dimension suggest different developmental mechanisms. For example, the study of free parental descriptions in seven countries found that in the Extraversion domain, the largest decrease in Sociability descriptors was evident from age 3 to 9 years and the largest decrease in Activity descriptors was evident from age 9 to 12 years (Slotboom et al., 1998). Soto et al. (2011) also found different age trends for the two facets of Extraversion, assertiveness and activity, as well as for the facets of Neuroticism, Conscientiousness and Openness, across adolescence. De Haan et al. (2017) showed that within the Extraversion domain, mean level of energy decreased, while mean levels of optimism and shyness did not change from age 2 to 5 years. Shyness is associated with Extraversion in the HiPIC and with Neuroticism in the ICID, however, studies using these instruments concur that shyness shows no change during preschool years (De Haan et al., 2017; Zupančič et al., 2009) and an increase from later childhood through adolescence (De Haan et al., 2017; Slobodskaya & Akhmetova, 2010). Although shyness, unlike unsociability or preference for solitary-constructive and exploratory behaviour, is associated with negative emotions (Chen, 2019; Rubin et al., 2009), its developmental trajectory is similar to those of Neuroticism and Extraversion and possibly indicates a common developmental mechanism.

It is notable, too, that a universal pattern of personality development in adulthood is characterised by increases in the traits within the Alpha domain, Agreeableness, Conscientiousness and Neuroticism (reversed), and decreases in the traits within the Beta domain, Extraversion and Openness (Costa Jr et al., 2019). There is considerable evidence on the neurobiological developmental processes associated with the two higher-order factors of the Big Five and the three temperament factors (Block, 2010; Rothbart, 2011; Shiner & DeYoung, 2013). Gender differences in age trends for some child and adolescent traits point to additional sources of influence, such as gender-specific socialisation or the timing of puberty (De Bolle et al., 2015; Klimstra et al., 2009; Van den Akker et al., 2014). Finally, cultural differences and similarities may help to delineate universal and culture-specific developmental processes leading to personality/temperament change (Slobodskaya, 2019).

Although research on the processes underlying personality

development in childhood and adolescence is still at an early stage, developments in this area will allow a better understanding of neurobiological processes and environmental factors associated with changes in child and adolescent traits.

6. Conclusions and directions

Research on the structure of child and adolescent personality suggests that the hierarchy of traits is broadly universal and incorporates well-established models of child temperament (Caspi et al., 2005; De Pauw, 2017; Shiner & DeYoung, 2013; Soto & John, 2014; Tackett et al., 2012). Future studies should consider similarities and differences in personality structure across informants, ages and cultures and integrate neurobiological processes and environmental factors associated with in temperament/personality traits. Developmental trends across the first ten years of life resemble well-documented maturational trends in adult personality – Agreeableness and Conscientiousness increase and Extraversion and Openness decline (Costa Jr et al., 2019); with the exception of Neuroticism, which tends to increase during childhood but declines in adulthood. There is evidence for disruption in personality maturation during adolescence, with mean levels of Agreeableness, Conscientiousness and Openness declining from late childhood through early adolescence and increasing from late adolescence through early adulthood (Denissen et al., 2013; Soto, 2016; Soto et al., 2011; Van den Akker et al., 2014), although the findings are inconsistent (Klimstra et al., 2009; Luan et al., 2017). Overall, our understanding of personality development in childhood and adolescence is still at an early stage (De Pauw, 2017; Tackett & Durbin, 2017).

Whether and to what extent the development of child and adolescent traits differs by gender remains an open question (De Haan et al., 2017; Soto, 2016). The findings suggest that personality maturation occurs earlier in girls and that girls, on average, tend to score increasingly higher on Neuroticism than boys during adolescence; however, the literature is equivocal. One explanation for inconsistent findings is that traits show curvilinear patterns of change including negative and positive age trends as well as periods of stability (Borghuis et al., 2017; Branje et al., 2007; Klimstra et al., 2009). Therefore, it may be necessary to estimate mean levels of traits year by year across childhood and adolescence (Soto et al., 2011). The Big Five traits have accumulated the most evidence on personality change (Borghuis et al., 2017; Branje et al., 2007; Costa Jr et al., 2019; Göllner et al., 2017; Lamb et al., 2002; Luan et al., 2017; Pullmann et al., 2006; Van den Akker et al., 2014). Examining age trends at different levels of the personality hierarchy is one approach to learning about the mechanisms leading to trait change. Research indicates that age and gender differences in child and adolescent personality traits may be observed differently by different raters (Branje et al., 2007; Göllner et al., 2017; Luan et al., 2017; Van den Akker et al., 2014). Multi-informant studies can help to better understand personality development. Inconsistent findings may also be attributable to contextual specificity; studies with children and adolescents from varying backgrounds may help to delineate the universal and community-specific age trends. Most of the research has been carried out in western countries (Slobodskaya, 2019), therefore, it is it is crucial to consider the development of child and adolescent personality in other cultures. More work on normative changes in child personality and individual differences in change is needed to understand the processes of the developmental change and predict associated outcomes.

Acknowledgements

This research was supported in part by the Russian Foundation for Basic Research grant # 19-013-00034 (data collection) and the Russian Science Foundation grant # 16-18-00003 (preparation of this manuscript).

References

Allik, J., Laidra, K., Realo, A., & Pullmann, H. (2004). Personality development from 12 to 18 years of age: Changes in mean-levels and structure of traits. *European Journal of Personality, 18*, 445–462.

Allik, J., Realo, A., Asendorpf, J. B., Baumert, A., Schmitt, M., Blum, G., … Mõttus, R. (2017). Open peer commentary and Authors' response. *European Journal of Personality, 31*, 529–595.

Baltes, P., Lindenberger, U., & Staudinger, U. (2006). Life span theory in developmental psychology. In W. Damon & R. Lerner (Eds.), Handbook of child psychology: Vol. 1. Theoretical models of human development (6th ed., pp. 569-595). New York: Wiley.

Baumert, A., Schmitt, M., Perugini, M., Johnson, W., Blum, G., Borkenau, P., … Jayawickreme, E. (2017). Integrating personality structure, personality process, and personality development. *European Journal of Personality, 31*, 503–528.

Block, J. (2010). The five-factor framing of personality and beyond: Some ruminations. *Psychological Inquiry, 21*, 2–25.

Borghuis, J., Denissen, J. J., Oberski, D., Sijtsma, K., Meeus, W. H., Branje, S., … Bleidorn, W. (2017). Big Five personality stability, change, and codevelopment across adolescence and early adulthood. *Journal of Personality and Social Psychology, 113*, 641–657.

Brandes, C. M., Reardon, K. W., & Tackett, J. L. (2020). Theories of personality. In S. Hupp, & J. Jewell (Eds.), *The Encyclopedia of Child and Adolescent Development*. John Wiley & Sons.

Branje, S. J. T., Van Lieshout, C. F. M., & Gerris, J. R. M. (2007). Big Five personality development in adolescence and adulthood. *European Journal of Personality, 21*, 45–62.

Bronfenbrenner, U. (1999). Environments in developmental perspective: Theoretical and operational models. In S. L. Friedman, & T. D. Wachs (Eds.), *Measuring environment across the life span: Emerging methods and concepts* (pp. 3–28). Washington, DC: American Psychological Association Press.

Caspi, A., Roberts, B. W., & Shiner, R. L. (2005). Personality development: Stability and change. *Annual Review of Psychology, 56*, 453–484.

Chen, X. (2019). Culture and shyness in childhood and adolescence. *New Ideas in Psychology, 53*, 58–66.

Cheung, F. M., van de Vijver, F. J., & Leong, F. T. (2011). Toward a new approach to the study of personality in culture. *American Psychologist, 66*, 593–603.

Costa, P. T., Jr., McCrae, R. R., & Löckenhoff, C. E. (2019). Personality across the life span. *Annual Review of Psychology, 70*, 423–448.

De Bolle, M., De Fruyt, F., McCrae, R. R., Löckenhoff, C. E., Costa, P. T., Jr., Aguilar-Vafaie, M. E., … Avdeyeva, T. V. (2015). The emergence of sex differences in personality traits in early adolescence: A cross-sectional, cross-cultural study. *Journal of Personality and Social Psychology, 108*, 171–185.

De Fruyt, F., De Bolle, M., McCrae, R. R., Terracciano, A., Costa, P. T., Jr., Ahn, C.-Y., … Shimonaka, Y. (2009). Assessing the universal structure of personality in early adolescence: The NEO-PI-R and NEO-PI-3 in 24 cultures. *Assessment, 16*, 301–311.

De Fruyt, F., Van Hiel, A., & Buyst, V. (1998). Parental personality descriptors of boys and girls. In G. A. Kohnstamm, C. F. Halverson, Jr., I. Mervielde, & V. L. Havill (Eds.), *Parental descriptions of child personality: Developmental antecedents of the big five?* (pp. 155–167). Hillsdale: Erlbaum, New Jersey.

De Haan, A., De Pauw, S., van den Akker, A., Deković, M., & Prinzie, P. (2017). Long-term developmental changes in children's lower-order big five personality facets. *Journal of Personality, 85*, 616–631.

De Pauw, S. S. (2017). Childhood personality and temperament. In T. A. Widiger (Ed.), *The Oxford handbook of the five-factor model* (pp. 243–280). NY: Oxford University Press.

De Pauw, S. S. W., Mervielde, I., & Van Leeuwen, K. G. (2009). How are traits related to problem behavior in preschoolers? Similarities and contrasts between temperament and personality. *Journal of Abnormal Child Psychology, 37*, 309–325.

De Raad, B., & Mlačič, B. (2017). Psycho-lexical studies of personality structure across cultures. In T. Church (Ed.), *The Praeger handbook of personality across cultures* (pp. 161–192). Santa Barbara: Praeger Publishers.

Denissen, J. J., van Aken, M. A., Penke, L., & Wood, D. (2013). Self-regulation underlies temperament and personality: An integrative developmental framework. *Child Development Perspectives, 7*, 255–260.

Eisenberg, N., Spinrad, T. L., & Knafo-Noam, A. (2015). Prosocial development (Vol. Ed.). (Series Ed.). In M. Lamb, & R. M. Lerner (Eds.), *Handbook of child psychology and developmental science: Vol. 3. Socioemotional processes* (7th ed., pp. 610–656). New York, NY: Wiley.

Else-Quest, N. M., Hyde, J. S., Goldsmith, H. H., & Van Hulle, C. A. (2006). Gender differences in temperament: A meta-analysis. *Psychological Bulletin, 132*, 33–72.

Göllner, R., Roberts, B. W., Damian, R. I., Lüdtke, O., Jonkmann, K., & Trautwein, U. (2017). Whose "storm and stress" is it? Parent and child reports of personality development in the transition to early adolescence. *Journal of Personality, 85*, 376–387.

Halverson, C. F., Havill, V. L., Deal, J., Baker, S. R., Victor, J., Pavlopoulos, V., … Wen, L. (2003). Personality structure as derived from parental ratings of free descriptions of children: The inventory of child individual differences. *Journal of Personality, 71*, 995–1026.

Halverson, C. F., Kohnstamm, G. A., & Martin, R. P. (Eds.). (1994). *The developing structure of temperament and personality from infancy to adulthood*. Hillsdale, NJ: Erlbaum.

Just, C. (2011). A review of literature on the general factor of personality. *Personality and Individual Differences, 50*, 765–771.

Kawamoto, T., & Endo, T. (2015). Personality change in adolescence: Results from a Japanese sample. *Journal of Research in Personality, 57*, 32–42.

Klimstra, T. (2013). Adolescent personality development and identity formation. *Child Development Perspectives, 7*, 80–84.
Klimstra, T. A., Hale, W. W., III, Raaijmakers, Q. A. W., Branje, S. J. T., & Meeus, W. H. J. (2009). Maturation of personality in adolescence. *Journal of Personality and Social Psychology, 96*, 898–912.
Knyazev, G. G., Zupancic, M., & Slobodskaya, H. R. (2008). Child personality in Slovenia and Russia: Structure and mean level of traits in parent and self-ratings. *Journal of Cross-Cultural Psychology, 39*, 317–334.
Kohnstamm, G. A., Halverson, C. F., Jr., Mervielde, I., & Havill, V. L. (Eds.). (1998). *Parental descriptions of child personality: Developmental antecedents of the big five?*. Mahwah, NJ: Erlbaum.
Laidra, K., De Fruyt, F., & Konstabel, K. (2017). Assessing childhood personality with the Estonian short version of the Hierarchical Personality Inventory for Children (HiPIC). *Personality and Individual Differences, 112*, 31–36.
Lamb, M. E., Chuang, S. S., Wessels, H., Broberg, A. G., & Hwang, C. (2002). Emergence and construct validation of the Big Five factors in early childhood: A longitudinal analysis of their ontogeny in Sweden. *Child Development, 73*, 1517–1524.
Lengua, L. J., Gartstein, M. A., & Prinzie, P. (2019). Temperament and personality trait development in the family: Interactions and transactions with parenting from infancy through adolescence. In D. P. McAdams, R. L. Shiner, & J. L. Tackett (Eds.), *Handbook of personality development* (pp. 201–220). New York, NY: Guilford Press.
Luan, Z., Hutteman, R., Denissen, J. J., Asendorpf, J. B., & van Aken, M. A. (2017). Do you see my growth? Two longitudinal studies on personality development from childhood to young adulthood from multiple perspectives. *Journal of Research in Personality, 67*, 44–60.
Markon, K. E., Krueger, R. F., & Watson, D. (2005). Delineating the structure of normal and abnormal personality: An integrative hierarchical approach. *Journal of Personality and Social Psychology, 88*, 139–157.
McAdams, D. P., & Olson, B. D (2010). Personality development: Continuity and change over the life course. *Annual Review of Psychology, 61*, 517–542.
McAdams, D. P., Shiner, R. L., & Tackett, J. L. (Eds.). (2019). *Handbook of personality development*. New York, NY: Guilford Press.
McCrae, R. R., Costa, P. T., Jr., Terracciano, A., Parker, W. D., Mills, C. J., De Fruyt, F., & Mervielde, I. (2002). Personality trait development from age 12 to age 18: Longitudinal, cross-sectional and cross-cultural analyses. *Journal of Personality and Social Psychology, 83*, 1456.
McCrae, R. R., Terracciano, A., De Fruyt, F., De Bolle, M., Gelfand, M. J., Costa, P. T., Jr., & 42 Collaborators of the Adolescent Personality Profiles of Cultures Project. (2010). The validity and structure of culture-level personality scores: Data from ratings of young adolescents. *Journal of Personality, 78*, 815–838.
Mervielde, I., & De Fruyt, F. (1999). Construction of the Hierarchical Personality Inventory for Children (HiPIC). In I. Mervielde, I. Deary, F. De Fruyt, & F. Ostendorf (Eds.), *Personality psychology in Europe. Proceedings of the Eight European Conference on personality psychology* (pp. 107–127). Tilburg, the Netherlands: Tilburg University Press.
Mroczek, D. K., & Little, T. D. (Eds.). (2006). *Handbook of personality development*. Mahwah, NJ: Erlbaum.
Prinzie, P., & Deković, M. (2008). Continuity and change of childhood personality characteristics through the lens of teachers. *Personality and Individual Differences, 45*, 82–88.
Pullmann, H., Raudsepp, L., & Allik, J. (2006). Stability and change in adolescents' personality: A longitudinal study. *European Journal of Personality, 20*, 447–459.
Rothbart, M. K. (2011). *Becoming who we are: Temperament and personality in development*. New York, NY: The Guilford Press.
Rubin, K. H., Coplan, R. J., & Bowker, J. C. (2009). Social withdrawal in childhood. *Annual Review of Psychology, 60*, 141–171.
Shiner, R. L., & DeYoung, C. G. (2013). The structure of temperament and personality traits: A developmental perspective. In P. Zelazo (Ed.), *Oxford handbook of developmental psychology* (pp. 113–141). New York: Oxford University Press.
Slobodskaya, H. R. (2019). Culture, context, and the development of traits. In D. P. McAdams, R. L. Shiner, & J. L. Tackett (Eds.), *Handbook of personality development* (pp. 221–236). New York, NY: Guilford Press.
Slobodskaya, H. R., & Akhmetova, O. A. (2010). Personality development and problem behavior in Russian children and adolescents. *International Journal of Behavioral Development, 34*, 441–451.
Slotboom, A.-M., Havill, V. L., Pavlopolus, V., & De Fruyt, F. (1998). Developmental changes in personality descriptions of children: A cross-national comparison of parental descriptions of children. In G. A. Kohnstamm, C. F. Halverson, Jr., I. Mervielde, & V. L. Havill (Eds.), *Parental descriptions of child personality: Developmental antecedents of the big five?* (pp. 155–168). Hillsdale: Erlbaum, New Jersey.
Soto, C. J. (2016). The Little six personality dimensions from early childhood to early adulthood: Mean-level age and gender differences in parents' reports. *Journal of Personality, 84*, 409–422.
Soto, C. J., & John, O. P. (2014). Traits in transition: The structure of parent-reported personality traits from early childhood to early adulthood. *Journal of Personality, 82*, 182–199.
Soto, C. J., John, O. P., Gosling, S. D., & Potter, J. (2008). The developmental psychometrics of big five self-reports: Acquiescence, factor structure, coherence, and differentiation from ages 10 to 20. *Journal of Personality and Social Psychology, 94*, 718–737.
Soto, C. J., John, O. P., Gosling, S. D., & Potter, J. (2011). Age differences in personality traits from 10 to 65: Big-Five domains and facets in a large cross-sectional sample. *Journal of Personality and Social Psychology, 100*, 330–348.
Specht, J. (Ed.). (2017). *Personality development across the lifespan*. Academic Press.
Tackett, J. L., & Durbin, C. E. (2017). Advances in research on youth personality: Introduction to the special issue. *Journal of Research in Personality, 67*, 1–2.
Tackett, J. L., Slobodskaya, H. R., Mar, R. A., Deal, J., Halverson, C. H., Jr., Baker, S. R., … Besevegis, E. (2012). The hierarchical structure of childhood personality in five countries: Continuity from early childhood to early adolescence. *Journal of Personality, 80*, 847–879.
Tremblay, R. E. (2010). Developmental origins of disruptive behaviour problems: The "original sin"hypothesis, epigenetics and their consequences for prevention. *Journal of Child Psychology and Psychiatry, 51*, 341–367.
Van den Akker, A. L., Deković, M., Asscher, J., & Prinzie, P. (2014). Mean-level personality development across childhood and adolescence: A temporary defiance of the maturity principle and bidirectional associations with parenting. *Journal of Personality and Social Psychology, 107*, 736.
van der Linden, D., Vreeke, L., & Muris, P. (2013). Don't be afraid of the general factor of personality (GFP): Its relationship with behavioral inhibition and anxiety symptoms in children. *Personality and Individual Differences, 54*, 367–371.
van Geert, P., & Steenbeek, H. (2005). Explaining after by before: Basic aspects of a dynamic systems approach to the study of development. *Developmental Review, 25*, 408–442.
Wood, D., Harms, P., & Vazire, S. (2010). Perceiver effects as projective tests: What your perceptions of others say about you. *Journal of Personality and Social Psychology, 99*, 174–190.
Zupančič, M., & Kavčič, T. (2005). Gender differences in personality through early childhood: A multi-informant perspective. *Horizons of Psychology, 14*, 11–38.
Zupančič, M., Slobodskaya, H. R., & Knyazev, G. G. (2008). Gender differences in the child/adolescent personality traits: Slovenes and Russians compared. *Horizons of Psychology, 17*(3), 43–63.
Zupančič, M., Sočan, G., & Kavčič, T. (2009). Consistency in adult reports on child personality over the pre-school years. *European Journal of Developmental Psychology, 6*, 455–480.